Stochastic Processes

Infinite Horizon MDP Problems

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Outline

1. Introduction
2. Stochastic Shortest Path Problem
3. Discounted Problem
4. Average Cost Problem

Infinite Horizon Problems

- Assumptions
  - The number of stages is infinite (in practice, a finite but very large number of stages)
  - The system is stationary
- Typical results
  - Easy-to-implement algorithms (Policy iteration, Value iteration, etc.)
  - Stationary optimal policies

Infinite Horizon Problems: Example

- Asset Selling
  - Objective:

$$\max_{\mu_0,\cdots,\mu_{N-1}} E\left[\sum_{k=0}^{N-1} g_k(x_k,u_k,w_k)\right]$$

  - Per-stage reward

$$g_k(x_k,u_k,w_k) = \begin{cases} (1+r)^{N-k}x_k & \text{if } x_k \neq T \text{ and } u_k = u^1, \\ 0 & \text{otherwise} \end{cases}$$

- Optimal stationary policy as $N \to \infty$

  accept the offer x_k if $x_k > \bar{\alpha}$

  reject the offer x_k if $x_k < \bar{\alpha}$

Infinite v.s. Finite

- Difficulty in modeling
  - The total expected cost may go to infinity

$$\lim_{N\to\infty} E\left[\sum_{k=0}^{N-1} g_k(x_k, u_k, w_k)\right]$$

- Difficulty in solution
  - DP algorithm for finite horizon problem does not apply directly
  - How to define a terminate stage and terminate cost to start
  - How to define the cost-to-go function

Modeling Infinite Horizon Problems

- Stochastic shortest path (SSP) problems
  - There is a special cost-free termination state
  - The termination is inevitable
  - The number of stages to terminate is random
- Discounted problems $(0 < \alpha < 1)$

$$\lim_{N\to\infty} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k) \right\}$$

- Average cost per stage problems

$$\lim_{N\to\infty} \frac{1}{N} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} g(x_k, \mu_k(x_k), w_k) \right\}$$

Preview of infinite horizon results

- Key issue: Exploit the relation between optimal cost function J^* of the infinite horizon problem and the optimal cost-to-go functions J_k of the corresponding N-stage (finite horizon) problems

Preview of infinite horizon results

- $J_N(x)$: optimal cost of the N-stage problem
- Generated by DP algorithm with $J_0(x) = 0$

$$J_{k+1}(x) = \min_{u \in U(x)} \underset{w}{E} \{g(x,u,w) + J_k(f(x,u,w))\}$$

- Typical Results
  - Convergence of $J_N(x)$ to J^*

$$J^*(x) \stackrel{\text{def}}{=} \min_{\pi} J_\pi(x) = \lim_{N \to \infty} J_N(x), \quad \forall x$$

  - Bellman's equation

$$J^*(x) = \min_{u \in U(x)} \underset{w}{E} \{g(x,u,w) + J^*(f(x,u,w))\}$$

  - Optimality condition: If $\mu(x)$ minimizes Bellman's Eq., $\{\mu, \mu, \ldots\}$ is optimal (stationary policy)

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Stochastic Shortest Path (SSP) Problem

- State space contains $1, \cdots, n$ and a cost-free termination state t
- Finite control space $u \in U(i)$ for all $i \in \{1, \cdots, n, t\}$
- State t satisfies $p_{tt}(u) = 1$ and $g(t, u) = 0$ for all $u \in U(t)$
  - Deterministic shortest path problem is the special case of SSP where $p_{ij}(u) = 1$ for a unique state j for each state-action pair (i, u)
- Total cost function:

$$J_\pi(i) = \lim_{N \to \infty} E\left[\sum_{k=0}^{N-1} g_k(x_k, \mu_k(x_k), w_k) \middle| x_0 = i\right]$$

  where $\pi = \{\mu_0, \mu_1, \cdots\}$
- Objective: $J^*(i) = \min_\pi J_\pi(i)$

Guarantee of termination

- Consider the problems where reaching the termination state is inevitable
- Assumption: There exists an integer m such that regardless of the policy used and the initial state, there is a positive probability that the termination state will be reached after no more than m stages

$$\rho_\pi = \max_{i=1,\ldots,n} P\{x_m \neq t | x_0 = i, \pi\} < 1, \quad \forall \pi$$

Main Results

- (a) Given any initial conditions $J_0(1), ..., J_0(n)$, the sequence $J_k(i)$ generated by DP algorithm

$$J_{k+1}(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J_k(j) \right], \forall i$$

  converges to the optimal cost $J^*(i)$, where $g(i,u) = E_w[g(i,u,w)]$
- (b) Following Bellman's equation has $J^*(i)$ as unique solution:

$$J^*(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J^*(j) \right], \forall i$$

Main Results

- (c) For any stationary policy μ, the costs $J_\mu(1), \cdots, J_\mu(n)$ are unique solution of the equation

$$J_\mu(i) = g(i, \mu(i)) + \sum_{j=1}^{n} p_{ij}(\mu(i)) J_\mu(j), \forall i$$

  Furthermore, given any initial conditions $J_0(1), ..., J_0(n)$, the sequence $J_k(i)$ generated by DP algorithm,

$$J_{k+1}(i) = g(i, \mu(i)) + \sum_{j=1}^{n} p_{ij}(\mu(i)) J_k(j), \forall i$$

  converges to the cost $J_\mu(i)$ for each i

- (d) A stationary policy μ is optimal if and only if for every state i, $\mu(i)$ attains the minimum in Bellman's equation

Proof of (a)

- Denote $\rho = \max_\pi \rho_\pi < 1$, we have

$$|J_\pi(i)| \le \sum_k m\rho^k \max|g| = \frac{m}{1-\rho}\max|g|$$

- For a given policy $\pi = \{\mu_0, \mu_1, \cdots\}$, rewrite

$$J_\pi(x_0) = \underbrace{E\left\{\sum_{k=0}^{mK-1} g(x_k, \mu_k(x_k))\right\}}_{\text{finite horizon with zero terminal cost}} + \underbrace{E\left\{\sum_{k=mK}^{\infty} g(x_k, \mu_k(x_k))\right\}}_{\text{tail cost}}$$

- The tail cost satisfies

$$\left|E\left\{\sum_{k=mK}^{\infty} g(x_k, \mu_k(x_k))\right\}\right| \le \frac{\rho^K}{1-\rho} m \max|g|$$

Proof of (a)

$$\left| J_\pi(x_0) - E\left\{ \sum_{k=0}^{mK-1} g(x_k, \mu_k(x_k)) \right\} \right| \leq \frac{\rho^K}{1-\rho} m \max |g|$$

- Taking the minimum over π we obtain

$$|J^*(x_0) - J_{mK}(x_0)| \leq \frac{\rho^K}{1-\rho} m \max |g|$$

where $J_{mK}(x_0)$ is the optimal cost for the mK-stage cost problem, generated by DP algorithm

- Taking $K \to \infty$, we obtain

$$\lim_{K\to\infty} J_{mK}(x_0) = J^*(x_0)$$

Spider and Fly

- A spider and a fly move along a straight line
- Each time period, the fly moves one unit left with prob. p, one right with prob. p, and stays still otherwise
- The spider knows the position of the fly at the period beginning
- The spider's strategy
  - move one unit towards the fly if distance > 1
  - either move one unit or stay still if distance $= 1$
  - capture (terminate) if in the same position
- Problem: Capture the fly in minimum expected time

Spider and Fly: Formulation

- State: distance between spider and fly $x_k \in \{0, 1, \cdots, n\}$
  - Initial state: $x_0 = n$
  - Terminal state: $x_k = 0$
- Action: $u_k = 1$ (move towards fly) if $x_k \geq 2$, $u_k \in \{0, 1\}$ if $x_k = 1$
- State transition:

$$p_{ii}(1) = p_{i(i-2)}(1) = p, p_{i(i-1)}(1) = 1 - 2p, \quad i \geq 2$$

$$p_{11}(1) = 2p, p_{10}(1) = p_{11}(0) = 1 - 2p, p_{12}(0) = p_{10}(0) = p$$

- Per-stage cost:

$$g(i, u) = 1, \forall i \neq 0$$

Spider and Fly: Solution

- Directly solve Bellman's Equation

$$J^*(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J^*(j) \right], \forall i$$

Algorithm I: Value Iteration

- Directly apply DP iteration

$$J_{k+1}(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J_k(j) \right], \forall i$$

  for any initial conditions, for instance $J_0(i) = 0$.
- It is guaranteed that $\lim_{k \to \infty} J_k(i) = J^*(i)$

Algorithm II: Policy Iteration

- Applying the following conclusion: For any stationary policy μ, the linear system of n equations

$$J_\mu(i)=g(i,\mu(i))+\sum_{j=1}^{n}p_{ij}(\mu(i))J_\mu(j), i=1,\ldots,n$$

  has a unique solution $J_\mu(i), i=1,\ldots,n$
- Policy iteration method:
  - Policy evaluation: Solve the above linear equations for μ^k to obtain $J_{\mu^k}(i), i=1,\ldots,n$
  - Policy improvement: Find an improved policy

$$\mu^{k+1}(i)=\arg\min_{u\in U(i)}\left[g(i,u)+\sum_{j=1}^{n}p_{ij}(u)J_{\mu^k}(j)\right], \forall i$$

  - Terminate condition: $J_{\mu^{k+1}}(i)=J_{\mu^k}(i)$ for all i

Main Results

- (e) The policy iteration given by

$$\mu^{k+1}(i) = \arg\min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J_{\mu^k}(j) \right], \forall i$$

generates an improving sequence of policies, that is,

$$J_{\mu^{k+1}}(i) \leq J_{\mu^k}(i), \quad \forall i, k$$

where $J_{\mu^k}(i)$ is the solution of

$$J_{\mu^k}(i) = g(i, \mu^k(i)) + \sum_{j=1}^{n} p_{ij}(\mu^k(i)) J_{\mu^k}(j),$$

and terminates with an optimal policy

Proof of (e)

- Consider the recursion for all i

$$J_0(i) = J_{\mu^k}(i)$$

$$J_{m+1}(i) = g(i, \mu^{k+1}(i)) + \sum_{j=1}^{n} p_{ij}(\mu^{k+1}(i)) J_m(j), m \geq 0$$

- Due to the optimality of μ^{k+1},

$$J_0(i) = g(i, \mu^k(i)) + \sum_{j=1}^{n} p_{ij}(\mu^k(i)) J_0(j)$$

$$\geq g(i, \mu^{k+1}(i)) + \sum_{j=1}^{n} p_{ij}(\mu^{k+1}(i)) J_0(j) = J_1(i)$$

Proof of (e)

- Then, following the recursion, we have

$$J_1(i) = g(i, \mu^{k+1}(i)) + \sum_{j=1}^{n} p_{ij}(\mu^{k+1}(i)) J_0(j)$$
$$\geq g(i, \mu^{k+1}(i)) + \sum_{j=1}^{n} p_{ij}(\mu^{k+1}(i)) J_1(j) = J_2(i) \geq J_3(i) \geq \cdots$$

- Since $J_m(i) \to J_{\mu^{k+1}}(i)$, we obtain

$$J_{\mu^k}(i) = J_0(i) \geq J_{\mu^{k+1}}(i)$$

- Since the number of stationary policies is finite, we have $J_{\mu^k}(i) = J_{\mu^{k+1}}(i)$ for some k
- $J_{\mu^k}(i)$ satisfies Bellman's Equation. Thus, $J_{\mu^k}(i) = J^*(i)$

Algorithm III: Linear Programming

- In the value iteration, select an initial vector $(J_0(1), \ldots, J_0(n))$ such that

$$J_0(i) \leq \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J_0(j) \right], \forall i$$

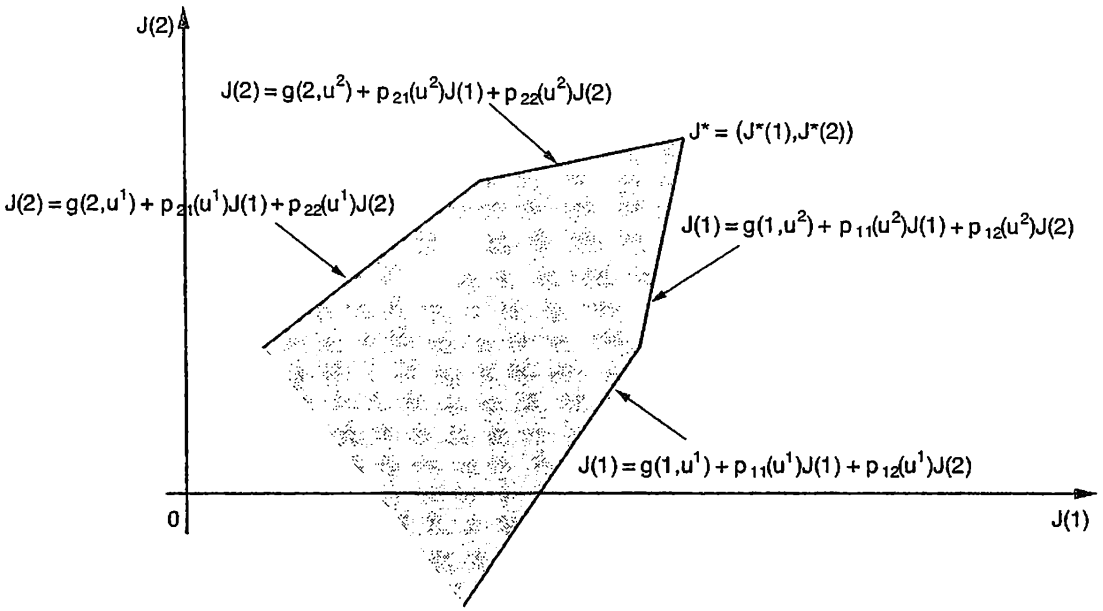
Then, we have $J_k(i) \leq J_{k+1}(i)$ for all k and i.
Since $J_k \to J^*$, J^* is the "largest" J

Algorithm III: Linear Programming

- $J^*(1), \ldots, J^*(n)$ solves the linear programming problem

$$\max_{J(1),\ldots,J(n)} \sum_{i=1}^{n} J(i)$$

$$\text{s.t. } J(i) \leq g(i,u) + \sum_{j=1}^{n} p_{ij}(u) J(j), \quad i = 1, \ldots, n$$



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Discounted Problem Formulation

$$\min_{\pi} \lim_{N\to\infty} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k) \right\}, \quad 0 < \alpha < 1$$

- Well-defined when g is bounded

$$\lim_{N\to\infty} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k) \right\}$$
$$\leq \lim_{N\to\infty} \sum_{k=0}^{N-1} \alpha^k \max|g| = \frac{\max|g|}{1-\alpha} < \infty$$

Unbounded Per-stage Cost Case

- When $g(x_k, \mu_k(x_k), w_k)$ is unbounded, but $|g| \leq Bk^n$,

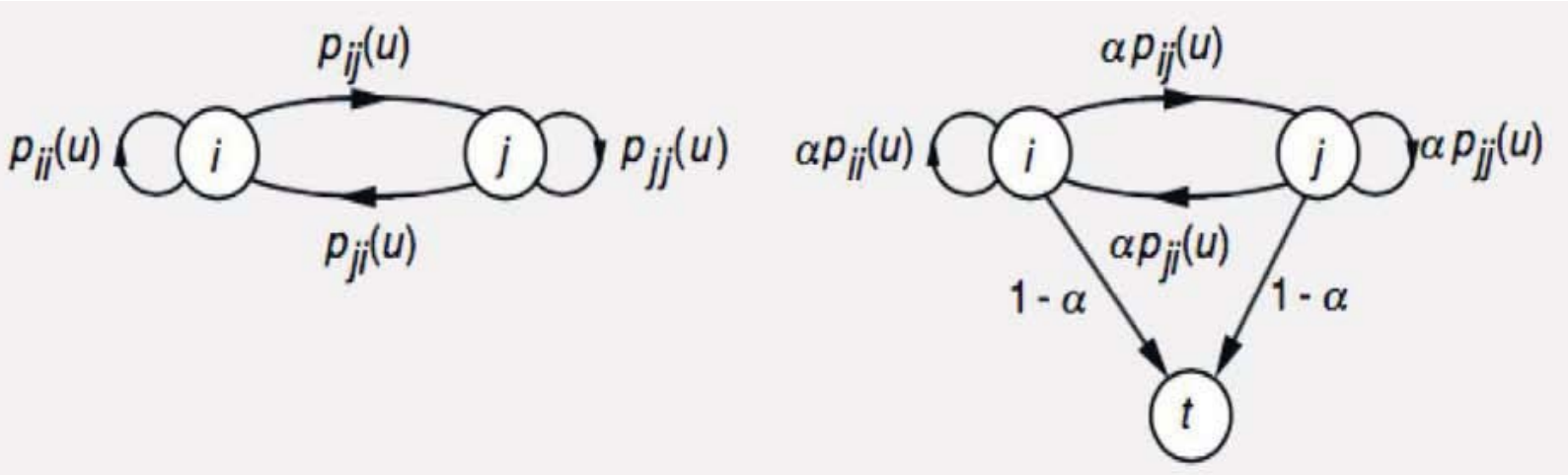
$$\lim_{N\to\infty} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k) \right\}$$
$$\leq \lim_{N\to\infty} \sum_{k=0}^{N-1} B\alpha^k k^n < \infty$$

  The discounted total cost is still well-defined

- For the general case that the discounted total cost goes to infinity under some policies, see Chapter 3 of Vol. II

Relation with SSP Problem

- Convert to an SSP problem



- k-th stage cost is $\alpha^k g(x_k, \mu_k(x_k))$, the same for both problems

Main Results

- (a) Given any initial conditions $J_0(1), ..., J_0(n)$, the sequence $J_k(i)$ generated by DP algorithm,

$$J_{k+1}(i) = \min_{u \in U(i)} \left[g(i,u) + \alpha \sum_{j=1}^{n} p_{ij}(u) J_k(j) \right], \forall i$$

  converges to the optimal cost $J^*(i)$ for each i
- (b) Bellman's equation has $J^*(i)$ as unique solution:

$$J^*(i) = \min_{u \in U(i)} \left[g(i,u) + \alpha \sum_{j=1}^{n} p_{ij}(u) J^*(j) \right], \forall i$$

Main Results

- (c) For any stationary policy μ, the costs $J_\mu(1), \cdots, J_\mu(n)$ are unique solution of the equation

$$J_\mu(i) = g(i, \mu(i)) + \alpha \sum_{j=1}^{n} p_{ij}(\mu(i)) J_\mu(j), \forall i$$

  Furthermore, given any initial conditions $J_0(1), ..., J_0(n)$, the sequence $J_k(i)$ generated by DP algorithm,

$$J_{k+1}(i) = g(i, \mu(i)) + \alpha \sum_{j=1}^{n} p_{ij}(\mu(i)) J_k(j), \forall i$$

  converges to the cost $J_\mu(i)$ for each i
- (d) A stationary policy μ is optimal if and only if for every state i, $\mu(i)$ attains the minimum in Bellman's equation

Main Results

- (e) The policy iteration algorithm given by

$$\mu^{k+1}(i) = \arg\min_{u \in U(i)} \left[g(i,u) + \alpha \sum_{j=1}^{n} p_{ij}(u) J_{\mu^k}(j) \right], \forall i$$

generates an improving sequence of policies and terminates with an optimal policy

Example: Batch Manufacturing

- A Manufacturer at each time period receives an order for her product with probability p and receives no order with probability $1-p$
- At any period, she has a choice of processing all the unfilled orders in a batch, or process no order at all. The maximum number of orders that can remain unfilled is n
- The cost per unfilled order at each time period is $c > 0$, the setup cost to process the unfilled orders is $K > 0$
- The manufacturer wants to find a processing policy that minimizes the total expected cost with discount factor $\alpha < 1$

Batch Manufacturing: Formulation

- State $i \in \{0, 1, \cdots, n\}$: number of unfilled orders
- Action $u \in \{0, 1\}$: process (1) or not (0)

$$u \in \{0, 1\}, \text{if } i < n; \quad u = 1, \text{if } i = n$$

- State Transition $p_{ij}(u)$:

$$p_{i1}(1) = p_{i(i+1)}(0) = p, \quad p_{i0}(1) = p_{ii}(0) = 1 - p, \quad i < n$$

$$p_{n1}(1) = p, \quad p_{n0}(1) = 1 - p$$

- Per-stage cost

$$g(i, 1) = K, \quad g(i, 0) = ci$$

Batch Manufacturing: Solution

- Value iteration

$$
\begin{aligned}
J_{k+1}(i) = \min \big[& K + \alpha(1-p)J_k(0) + \alpha p J_k(1), \\
& ci + \alpha(1-p)J_k(i) + \alpha p J_k(i+1) \big], \quad i = 0, 1, \cdots, n-1, \\
J_{k+1}(n) = & K + \alpha(1-p)J_k(0) + \alpha p J_k(1)
\end{aligned}
$$

- Convergence:

$$J^*(i) = \lim_{k \to \infty} J_k(i)$$

Batch Manufacturing: Property

- Bellman's equation

$$J^*(i) = \min\Big[K + \alpha(1-p)J^*(0) + \alpha p J^*(1),$$
$$ci + \alpha(1-p)J^*(i) + \alpha p J^*(i+1)\Big], \quad i = 0, 1, \cdots, n-1,$$
$$J^*(n) = K + \alpha(1-p)J^*(0) + \alpha p J^*(1)$$

- Monotonicity

$$J^*(i) \leq J^*(i+1)$$

- Proof
  - Prove by induction that $J_k(i) \leq J_k(i+1)$
  - Monotonicity holds for $J^*(i)$ since $J^*(i) = \lim_{k\to\infty} J_k(i)$

Batch Manufacturing: Threshold Policy

- Since $J^*(i)$ is nondecreasing in i, $ci + \alpha(1-p)J^*(i) + \alpha p J^*(i+1)$ is increasing in i
- There exist a threshold m such that

$$\begin{aligned} & c(m-1) + \alpha(1-p)J^*(m-1) + \alpha p J^*(m) \\ \leq\ & K + \alpha(1-p)J^*(0) + \alpha p J^*(1) \\ \leq\ & cm + \alpha(1-p)J^*(m) + \alpha p J^*(m+1) \end{aligned}$$

- The optimal threshold policy: to process when $i \geq m$, and not to process otherwise

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Average Cost Per Stage Problem

- Consider a stationary system with finite number of states and controls.
- Objective: minimizing

$$J_\pi(x_0) = \lim_{N\to\infty} \frac{1}{N} \underset{\substack{w_k \\ k=0,1,\ldots}}{E} \left\{ \sum_{k=0}^{N-1} g(x_k, \mu_k(x_k), w_k) \right\}$$

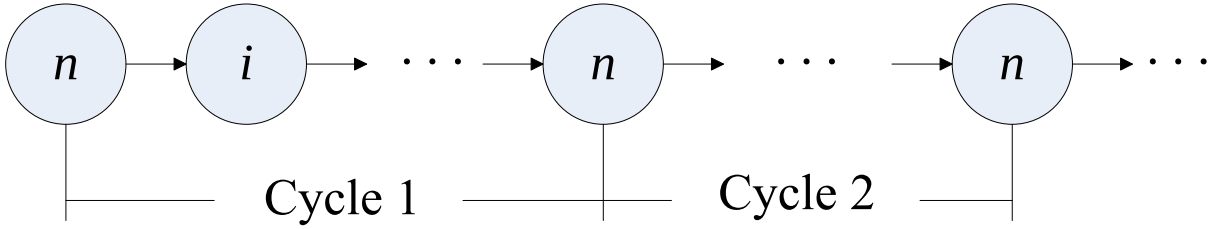
over policies $\pi = \{\mu_1, \mu_2, \ldots\}$

Important characteristics

- Given a stationary policy, a finite state Markov chain is formed
  - There are possibly multiple recurrent classes
  - Consider single recurrent class case
- For a single recurrent class, the cost incurred up to any fixed time T does not matter
- The optimal cost is independent of initial state, i.e., $J^*(i) = \lambda^*, \forall i$

Connection with SSP

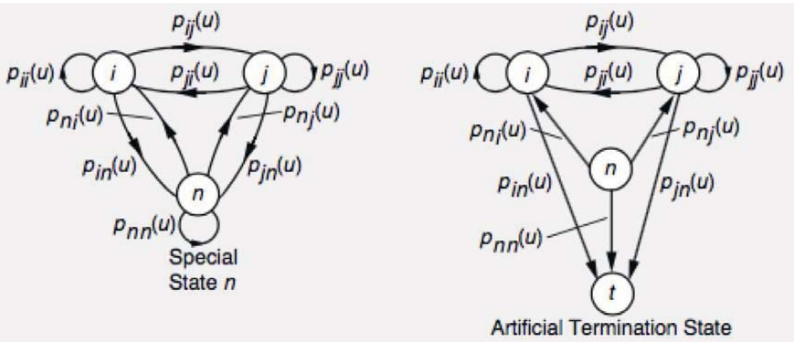
- Simplest case: One of the states n will be visited infinitely often (recurrent) for all initial states and all policies
- Divide the sequence of generated states into cycles marked by successive visits to n



  - The kth cycle includes the transitions from the $(k-1)$th to the kth visit to state n
  - The cycles are "independent" with each other
  - Each cycle can be viewed as a state trajectory of an SSP problem with n as the termination state

Connection with SSP

- Focus on a single cycle
  - Re-denote the termination state n as t, different from the initial state n
  - Equivalent to an SSP problem with t as the termination state



- Minimum average cost = minimum average cost of a cycle = minimum cost of a SSP

Connection with SSP

- Consider a minimum cycle cost problem: Find a stationary policy μ that minimizes the expected cost per transition within a cycle

$$\frac{C_{nn}(\mu)}{N_{nn}(\mu)} = \frac{\text{cost from } n \text{ up to the first return to } n}{\text{time from } n \text{ up to the first return to } n}$$

- Intuitively, $C_{nn}(\mu)/N_{nn}(\mu)$ is the average cost of μ, so $C_{nn}(\mu)/N_{nn}(\mu) \geq \lambda^*$. Therefore,

$$C_{nn}(\mu) - N_{nn}(\mu)\lambda^* \geq 0$$

- Minimizing $C_{nn}(\mu) - N_{nn}(\mu)\lambda^*$ is a SSP problem with per-stage cost $g(i,u) - \lambda^*$.
- If μ^* is optimal, $C_{nn}(\mu^*) - N_{nn}(\mu^*)\lambda^* = 0$ since λ^* is achievable.

Bellman's Equation

- Let $h^*(i)$ the optimal cost of this SSP problem when starting at the nontermination states $i = 1, ..., n$. Then $h^*(1), ..., h^*(n)$ solve uniquely the corresponding Bellman's equation

$$h^*(i) = \min_{u \in U(i)} \left[g(i,u) - \lambda^* + \sum_{j=1}^{n} p_{ij}(u) h^*(j) \right], \forall i$$

and

$$h^*(n) = C_{nn}(\mu^*) - N_{nn}(\mu^*)\lambda^* = 0$$

Bellman's Equation

- Or

$$\lambda^* + h^*(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) h^*(j) \right], \forall i$$

$$h^*(n) = 0$$

- Interpretation of $h^*(i)$ as a relative or differential cost: It is the minimum of $E\{$ cost to reach n from i for the first time $\}$ - $E\{$ cost if the stage cost were λ^* and not $g(i,u)$ $\}$

Main Results

- (a) The optimal average cost λ^* is the same for all initial states, and together with some vector $h^* = \{h^*(1), \cdots, h^*(n)\}$ satisfies Bellman's equation

$$\lambda^* + h^*(i) = \min_{u \in U(i)} \Big[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) h^*(j) \Big], \forall i$$

Furthermore, the stationary policy $\mu(i)$ is optimal if it attains the above minimum for all i. In addition, there is a unique vector h^* with $h^*(n) = 0$

Main Results

- (b) If a scalar λ and a vector $h = \{h(1), \cdots, h(n)\}$ satisfies Bellman's equation, then λ is the optimal average cost for all initial states
- (c) Given a stationary policy μ, the average cost λ_μ satisfies

$$\lambda_\mu + h_\mu(i) = g(i, \mu(i)) + \sum_{j=1}^{n} p_{ij}(\mu(i)) h_\mu(j), \forall i$$

where the vector $h_\mu = \{h_\mu(1), \cdots, h_\mu(n)\}$ is unique if $h_\mu(n) = 0$

Algorithms: Value Iteration

$$\lambda^* + h^*(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) h^*(j) \right], \forall i$$

- Unknown parameters $\lambda^*, h^*(i)$
- Relative value iteration
  Given $h_k(i)$, and for some fixed state s, calculate

$$\lambda_k = \min_{u \in U(s)} \left[g(s,u) + \sum_{j=1}^{n} p_{sj}(u) h_k(j) \right]$$

and

$$h_{k+1}(i) = \min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) h_k(j) \right] - \lambda_k$$

Algorithms: Policy Iteration

- At iteration k, we have a stationary μ^k
- Policy iteration: Solving linear equations

$$\lambda^k + h^k(i) = g(i, \mu^k(i)) + \sum_{j=1}^{n} p_{ij}(\mu^k(i)) h^k(j), \forall i$$

$$h^k(n) = 0$$

- Policy improvement:

$$\mu^{k+1}(i) = \arg\min_{u \in U(i)} \left[g(i,u) + \sum_{j=1}^{n} p_{ij}(u) h^k(j) \right], \forall i$$

- Terminate when $\lambda^{k+1} = \lambda^k, h^{k+1}(i) = h^k(i), \forall i$

Batch Manufacturing

- State $i \in \{0, 1, \cdots, n\}$: number of unfilled orders
- Action $u \in \{0, 1\}$: process (1) or not (0)

$$u \in \{0, 1\}, \text{if } i < n; \quad u = 1, \text{if } i = n$$

- State Transition $p_{ij}(u)$:

$$p_{i1}(1) = p_{i(i+1)}(0) = p, \quad p_{i0}(1) = p_{ii}(0) = 1 - p, \quad i < n$$

$$p_{n1}(1) = p, \quad p_{n0}(1) = 1 - p$$

- Per-stage cost

$$g(i, 1) = K, \quad g(i, 0) = ci$$

Batch Manufacturing: Average Cost Version

- Bellman's Equation

$$\begin{aligned}\lambda^* + h^*(i) &= \min\big[K + (1-p)h^*(0) + ph^*(1), \\ &\qquad ci + (1-p)h^*(i) + ph^*(i+1)\big], \quad i = 0, 1, \cdots, n-1, \\ \lambda^* + h^*(n) &= K + (1-p)h^*(0) + ph^*(1)\end{aligned}$$

Batch Manufacturing: Optimal Policy

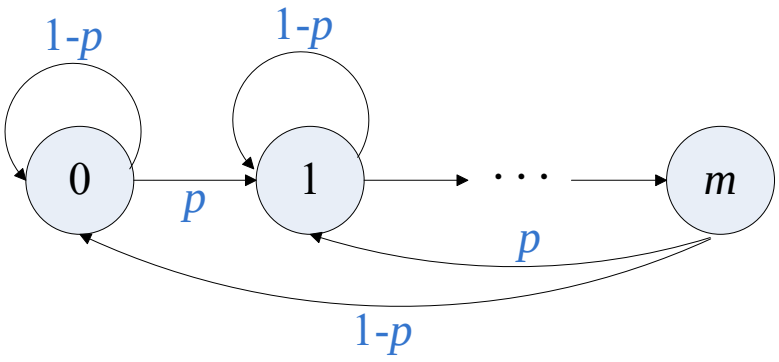
- The optimal policy is to process i unfilled orders if

$$K + (1-p)h^*(0) + ph^*(1) \leq ci + (1-p)h^*(i) + ph^*(i+1)$$

- Observation: $h^*(i) \leq h^*(i+1)$
  - Proof: Consider value iteration of the associate SSP problem
- The monotonicity implies the threshold policy is optimal

Batch Manufacturing: Solution

- Given a threshold m, a Markov chain is obtained



- By solving the stationary distribution $\boldsymbol{\pi}$, the average cost is

$$\sum_{i=0}^{m-1} ci\pi_i + K\pi_m := f(m)$$

- The optimal threshold can be obtained by minimizing $f(m)$

Project

- Write programs to solve batch manufacturing problem
  - Formulate and solve both discounted and average cost problems
  - Realize both value iteration and policy iteration algorithms
  - The parameters c, K, n, p, α as inputs
- Experiments and analysis
  - Run the code with different parameter inputs to show the performance of value iteration and policy iteration
  - For discounted problem, consider exercise 7.8
  - For average cost problem, write code to find the threshold by stationary analysis
  - Compare the optimal policies of the discounted problem with different values of α and the average cost problem
- Submit a report by Dec. 31, 23:59:59, 2022 (include source code in appendix)